ECG Data Analysis with IoT and Machine Learning

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***Abstract* - Regularity and irregularity in heartbeat rhythms are widely diagnosed through the analysis of electrocardiogram (ECG) recordings of the heart. However, clinical analysis comes with financial and scheduling costs that may not be necessary in the absence of heart illnesses. The goal of this research is to investigate if we can design a mobile health system that allows users to monitor and analyze their heart rhythms by using Internet of Things (IoT) techniques and machine learning methods. We proposed to have an IoT-based system embedded with a trained machine learning model to address this issue. The proposed system contains a few parts. An AD8232 sensor was used to collect ECG signals, which were then sent to a minicomputer, Raspberry Pi, through an analog-to-digital (ADC) converter. The digital ECG signals were fed to a Flask application and then sent to the machine learning model for classification. The machine learning model was trained on three different heart rhythms from the PhysioNet, including Normal Sinus Rhythm, Congestive Heart Failure, and Atrial Fibrillation. The training and testing processes were also in the IoT device. To support interaction for the public, a mobile web application was created to show the result of the classification as well as presenting real-time ECG data as a continuous graph. The result of the work demonstrates that IoT techniques not only allowed users to monitor their heartbeats, but it could also analyze real-time ECG signals with a trained machine learning model for anomaly detection.**

Keywords – electrocardiogram (ECG), machine learning, Raspberry Pi, Flask, anomaly detection, deep learning

I. Introduction

Electrocardiogram (ECG) recordings of the heart demonstrate patterns that distinguish regular heart rhythms from irregular heart rhythms. This kind of rhythm can be further used to diagnose heart conditions by medical professionals. Inspired by the fact that people with heart illness might have different patterns in ECG recordings than healthy people, we were motivated to design a new mobile health system that can adopt a trained machine learning model to classify different heart conditions.

In fact, the process of visiting a medical professional and doing follow-up extensive tests to diagnose illnesses have heavy financial and scheduling costs for users. Instead, there is a need to have a low-cost, easy-to-use heart monitoring device in a home setting. Given such kind of device, users can continuously monitor their hearts, through trained machine learning models to identify abnormal scenarios. Only when the anomaly condition is detected, users can contact their medical provider for follow up.

Instead of using a microcontroller-based platform, such as Arduino, which has limited computing power to run and deploy machine learning models and web applications, we decided to use a minicomputer, Raspberry Pi 4, to realize our idea.

In this paper, we will start from presenting related work, proposed idea, and hypothesis. Then, follow through with the design and implementation of the proposed system. The paper will be wrapped with discussion, conclusions, and future work.

II. Related Work

In terms of building a model for classifying ECG signals, researchers have produced machine learning and deep learning models in the past [1, 2]. Such models were developed to streamline the process of anomaly detection in the heart. Our research focuses on a mobile platform, which normally has limited computing power and resources.

To train a model that accurately classifies heartbeat type, we used ECG data from the PhysioNet, an online database for widely accepted medical data. The PhysioNet data has been extensively used for research and studies related to biological, physiological, and medical developments. In [3], Goldberger et al. stated that data stored in the PhysioNet was a substitute for conducting experiments to collect new data since regulated, pre-recorded data is available through the PhysioNet. Data presented in the PhysioNet replaces the need to distribute resources to gather quality data.

In terms of successful classification methods, our previous work [4] studied ECG data analysis with Multiscale Entropy (MSE) and seven different traditional machine learning methods. We found that Random Forest classifier can provide the best prediction result (86%) among all seven methods by dividing RR interval values with the sampling rate twice and have a unit of squared seconds per sample. Tran et al. [1] proposed a deep learning technique to classify ECG signals collected from IoT devices by using an Auto Encoder (AE) and a combination of a Long Short-Term Memory (LSTM) model with a Fully Connected (FC) layer and produced approximately 90% accuracy and F1 score. While Tran et al.'s work extracted RR Interval from ECG signals for analysis, Zhao et al. [2] proposed to use a deep learning approach to analyze raw ECG signals for arrhythmia detection. Specifically, Zhao et al. used a Convolutional Neural Network (CNN) based architecture to address the issue. This approach is also applicable to mobile telehealth systems.

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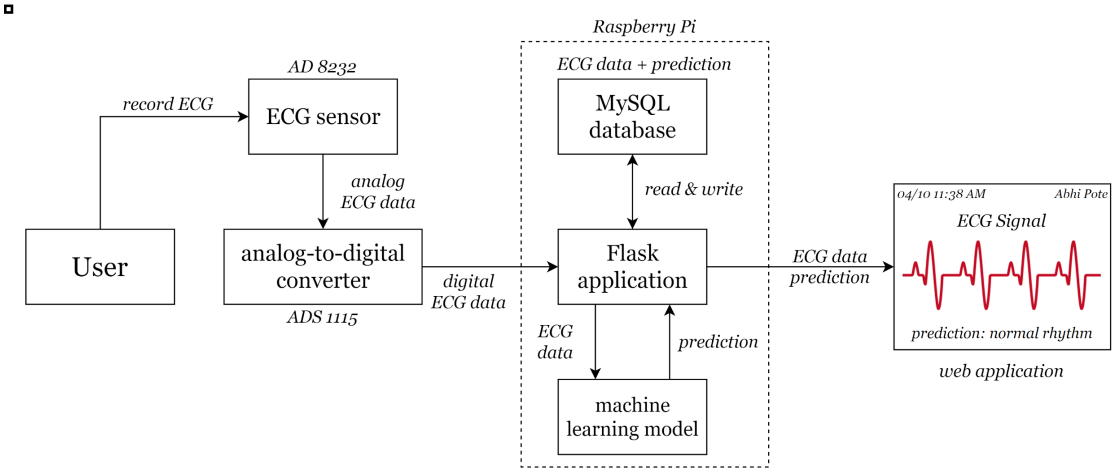

Figure 1. ECG data is collected from the user through the AD8232 sensor, which passes that data to the Raspberry Pi via the ADS1115 converter. Once the data reaches the Flask application in the Raspberry Pi, a machine learning model classifies it, and that information is stored in a MySQL database. Lastly, a web application displays the stored information.

III. Proposed Idea and Hypothesis

In this research, we focused on designing a new mobile health system that allows users to monitor and analyze their heartbeat data in real time. Considering the financial and scheduling costs of visiting a clinic and recording heartbeat for understanding heart rhythm, which normally is analyzed by experts, we hope to streamline this process with Internet of Things (IoT) techniques and machine learning methods. We hypothesize that the proposed system can monitor heart rhythms and identify anomaly in a home setting. When anomaly is detected, users can visit a clinic to confirm the detection and take necessary steps to protect their health.

IV. System Design

The design of the system includes software and hardware parts. In this section, we would like to explain our design in each of them with details.

4.1 Hardware Design

Since we expected the proposed system to be lightweight and provide an easy-to-use interface for the users, we considered to use a minicomputer rather than a microcontroller-based platform in the design. We chose Raspberry Pi, a low cost, credit-card sized computer, to prototype the system.

The ECG sensor used in recording real time ECG data was AD8232 single lead heart rate monitor designed by SparkFun [5]. According to the description via SparkFun, since ECG data is noisy, the AD8232 acts as an op amp to record a clear signal from the PR and QT Intervals easily [5]. The ECG data collected by the sensor has an analog output, but the Raspberry Pi reads only digital signals. To bridge the gap, we used the ADS1115 analog-to-digital converter [6] to convert signals from analog to digital for processing. The ADS1115 has 16-bit precision at 860 samples/second over I2C. The chip can be used as 4 single-ended input channels, but we only use one input coming from the ECG sensor.

In the prototype system, the converted digital ECG signals were received by a Flask application running on the Raspberry Pi. Flask is a micro web framework written in Python [11] and our system used it for providing a back-end service. The ECG data was sent to the machine learning module where a trained model makes a prediction based on the type of ECG data and sends it back to the Flask application. Then, the Flask application sent the ECG data and prediction of the heartbeat type to the MySQL database for storage. When a user made a request to view their ECG recording and heartbeat type, the Flask application fetched the information from the database and sent it back to the user. Figure 1 illustrates a flow chart of getting ECG data from the user to the Raspberry Pi and presenting the predicted results to the user on a mobile web interface.

To bridge the gap between the ECG sensor (AD8232) and the Raspberry Pi, the A0 input of the ADS1115 received signals from the output of the AD8232, as shown in Figure 2. The SDA pin of the ADS1115 is connected to the SDA input of the Raspberry Pi and the SCL pin of the ADS1115 is connected to the SCL input of the Raspberry Pi. This arrangement is placed on a breadboard to allow more space for multiple connections, including the ground (GND) and 3.3V connections for each device.

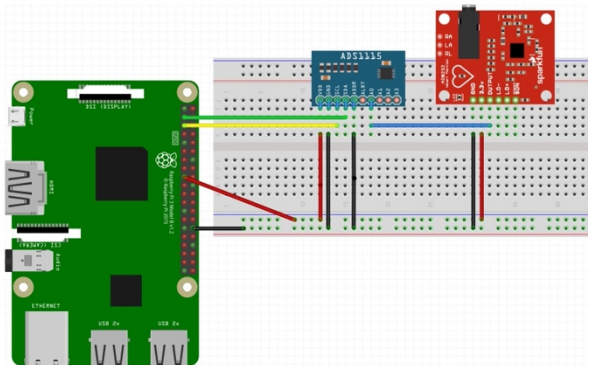

Figure 2. The diagram shows signals sent from the ECG sensor (red/right) to the Raspberry Pi (green/left) through the analog-to-digital converter (blue/middle), over a breadboard.

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4.2 Software Design

To consider the simplicity of the design, we decided to use Flask, a micro web framework written in Python, to design the back-end service with the MySQL relational database. Under such design, the system can not only deploy the trained machine learning model to predict the type of ECG signals, but it can also be scaled up to run the training and testing processes to have different trained models for extension.

V. Implementation

5.1 Data Collection

The ECG data was recorded through the AD8232 sensor which comes with three biomedical pads attached to the user's right arm, left arm, and right leg as illustrated in Figure 3. Since the AD8232 sensor recorded only analog data, we needed to send the analog signals to the ADS1115 analog-to-digital converter and send the digital output to the Raspberry Pi.

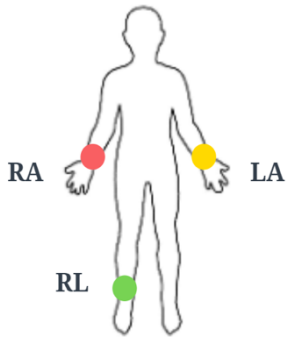

Figure 3. The sensor comes with three biomedical pads. The red pad is placed on the right arm (RA), yellow pad on the left arm (LA), and green pad on the right leg (RL).

5.2 Model Training and Testing

To train a machine learning model for testing ECG signals, we used datasets from the PhysioNet [3]. ECG data for subjects with Normal Sinus Rhythm (MIT-BIH NSR) [3], Congestive Heart Failure (BIDMC CHF) [7], and Atrial Fibrillation (MIT-BIH AF) [8] were extracted from the PhysioNet using the wfdb-python package [9]. In a similar study that dealt with abnormal heart rhythms, the records 04963 and 05091 in the MIT-BIH dataset had inaccurate annotations [10]. Likewise, subjects chf02 and chf06 in the BIDMC CHF dataset had high PAC or PVC frequency and were excluded from the study [10]. We removed the above mentioned four records from our study and divided the data into an equal number of segments for each heartbeat type. In total, 500 segments were created for each heartbeat type. Each segment contained 20,000 data points and a label was attached to each segment to indicate the heartbeat type. We randomized and split the dataset for training (70%) and testing (30%) respectively for training a machine learning model. Our previous work [3] found that Random Forest performed the best in ECG data prediction among seven machine learning methods. Therefore, we decided to train a Random Forest model for this research. The trained model was then used to predict the heartbeat type of the real-time ECG signals collected from the Raspberry Pi with the ECG sensor.

5.3 Data Analysis and Storage

The main Flask application within the Raspberry Pi received digital ECG signals from the ADS1115. The application received 20,000 ECG values during every instance of recording, which is considered as one segment of data. Each segment was passed from the main application to the machine learning model which predicts the heartbeat type of the incoming ECG signal. This prediction was sent back to the main application, which stored this prediction, corresponding segment of ECG data, and current date and time into the MySQL database. (Figure 1)

5.4 Mobile Web Application

To provide a reliable user interface and user experience, we designed an easy-to-use mobile web application that has buttons to record ECG signals and view heartbeat prediction. Upon recording ECG data and storing information to the database, the user is presented with their latest prediction result which involves a continuously moving graph of the most recent ECG recording, the prediction made by the trained machine learning model, and the date and time the data was recorded. (Figure 4) If any of the recent predictions do not indicate normal heartbeat rhythm, the user will be alerted immediately.

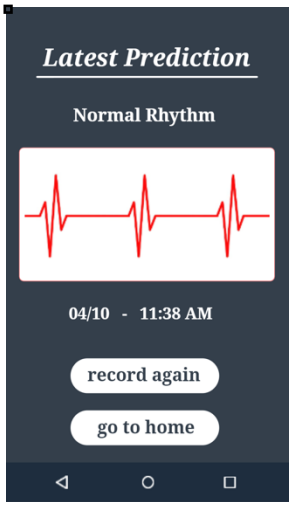

Figure 4. An easy-to-use mobile web application illustrates the user's ECG recording, heartbeat type, and date/time of the recording.

VI. Evaluation

Since the machine learning model was trained on segments with 20,000 ECG values, we also segmented every 20,000 ECG values of incoming signal to the Raspberry Pi as a segment. It took approximately 61 seconds to record one segment from the sensor before sending it to the trained model for prediction. Therefore, the user is required to hold

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the biomedical pads for just 1 minute to record a segment of ECG data and view a prediction of their heartbeat type.

To understand the effect of segment size with the prediction, we conducted an experiment on the model training process. We found that, upon training the model with 10,000 data points per segment, i.e., decreasing from original 20,000 data points per segment, the prediction score increased from 80% to 84%. In terms of the time to get the required data points, capturing 10,000 ECG values dropped from 61 seconds to 31 seconds. Likewise, the prediction score increased to 85% with 5,000 data points per segment and the time taken to record one segment dropped to 16 seconds, shown in Table 1. However, we used the original model, i.e., having 20,000 data points per segment, in the implementation for the prototype system.

Table 1. Comparison of different numbers of data points per segment for training and testing ECG signals.

Model No.	Data Points per Segment	Prediction Score	Time to Record One Segment (reference)
A	20,000	80%	61 seconds
B	10,000	84%	31 seconds
C	5,000	85%	16 seconds

VII. Discussion

As reported in the evaluation section, we found that the trained machine learning model can obtain 80% in accuracy during the training and testing process, and the evaluated ECG data came from the PhysioNet [3]. We expected this kind of trained model can be used in real life and embedded it to the IoT device with ECG sensor and provided a mobile web application as an interface. The prototype system has demonstrated our success in making a personal health system for ECG data analysis.

However, we found some potential issues during this process. For example, if the ECG sensor is not in a stable situation, i.e., the user does not stay still, or the biomedical pads loosen up while recording, the ECG data received by the Raspberry Pi might deviate from its actual value, and the classification made by the trained model might not be accurate. To overcome this issue, firm and reliable biomedical pads are required.

In addition, the machine learning method used in the Raspberry Pi was trained by three heartbeat datasets – Normal Sinus Rhythm, Congestive Heart Failure and Atrial Fibrillation. Although the results are satisfactory for this work, it does not tackle other heart conditions. Extensive datasets and research are needed to build models for unexplored heart conditions.

We found some studies utilized deep learning techniques to classify ECG data. Even though these models have a better accuracy than the one we presented here, the needed computation power is not suitable for Raspberry Pi. To benefit from deep learning approaches, more powerful IoT platforms need to be explored further, such as Raspberry Pi + USB Accelerator, Google Coral [12], NVIDIA Jetson platforms [13], etc.

VIII. Conclusions and Future Work

In this paper, we present a prototype system that allows users to monitor their heartbeat rhythms by using ECG signals with an IoT device, which can predict the user's heart condition within a minute. The prototype system used Raspberry Pi as a platform and an external ECG sensor to record real-time ECG signals from the user. In addition to showing real-time ECG signals as a continuous graph, the trained machine learning model was also able to classify the user's heartbeat rhythm and presented the result in a mobile web application. The prototype system used Random Forest method to train the machine learning model because we found it performed the best in ECG data analysis among other traditional machine learning methods, including decision tree, SVM, logistic regression, naïve bayes model, etc. [4] The goal of this research is not to replace any clinical visit, but instead is to allow self-diagnosis available at home and seek medical support in case of anomaly detection.

Although the prototype system has successfully demonstrated the idea of training and testing a machine learning model with an IoT device for classifying real-time ECG data, there are still some limitations. For example, the data used in training and testing is different from the data used in prediction. In fact, training and testing datasets from the PhysioNet might have come from hospitals with reduced noise levels, but real-time ECG signals recorded by the IoT device might face unexpected problems, such as misposition of the pad, unstable recording, increased noise etc., that might impact the prediction results. Lastly, in addition to traditional machine learning methods, we believe different deep learning approaches might provide an alternative solution to improve the accuracy of the prediction, but this needs more computation power in IoT devices. All of the topics mentioned above are important to investigate further to ensure personal medical devices can provide better anomaly detection.

References

[1] Duy T. Tran, Hung T. Vo, Dung D. Dguyen, Quan M. Nguyen, Liem T. Huynh, Ly T. Le, Hai T. Do and Tho T. Quan. "A Predictive Model for ECG Signals Collected from Specialized IoT Devices using Deep Learning." *2018 5th NAFOSTED Conference on Information and Computer Science (NICS)*, 2018.

[2] Peng Zhao, Dekui Quan, Wei Yu, Xinyu Yang and Xinwen Fu. "Towards Deep Learning-Based Detection Scheme with Raw ECG Signal for Wearable Telehealth Systems," *2019 28th International Conference on Computer Communication and Networks (ICCCN)*, 2019.

[3] Ary L. Goldberger, Luis A. N. Amaral, Leon Glass, Jeffrey M. Hausdorff, Plamen Ch. Ivanov, Roger G. Mark, Joseph E. Mietus, Beorge B. Moody, Chung-Kang Peng, and H. Eugene Stanley. "PhysioBank,

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PhysioToolkit, and PhysioNet: Components of a new research resource for complex physiologic signals." *Circulation*, 2000.

[4] Abhi Pote Shrestha and Chen-Hsiang Yu. "ECG Data Analysis with MSE and Machine Learning." *The 2021 IEEE MIT Undergraduate Research Technology Conference (IEEE URTC)*, October 2021.

[5] SparkFun Single Lead Heart Rate Monitor - AD8232. https://www.sparkfun.com/products/12650 (Accessed on December 25, 2021)

[6] ADS1115 16-Bit ADC - 4 Channel with Programmable Gain Amplifier. https://www.adafruit.com/product/1085 (Accessed on December 25, 2021)

[7] Donald S. Baim, Wilson S. Colucci, E. Scott Monrad, Harton S. Smith, Richard F. Wright, Alyce Lanoue, Diane F. Gauthier, Bernard J. Ransil, William Grossman and Eugene Braunwald. "Survival of patients with severe congestive heart failure treated with oral milrinone." *Journal of the American College of Cardiology*, 7(3), pp.661-670, March 1986.

[8] George B. Moody and Roger G. Mark. "A new method for detecting atrial fibrillation using R-R intervals." *Computers in Cardiology*, 1983.

[9] The native Python waveform-database (WFDB) package. MIT-LCP: wfdb-python. https://github.com/MIT-LCP/wfdb-python (Accessed on December 25, 2021)

[10] Kwon H. Yoon, Tharoeun Thap, Chang W. Jeong, Nam H. Kim, Seung Noh, Yunyoung Nam and Jinseok Lee. "Analysis of Statistical Methods for Automatic Detection of Congestive Heart Failure and Atrial Fibrillation with Short RR Interval Time Series." *2015 9th International Conference on Innovative Mobile and Internet Services in Ubiquitous Computing*, 2015

[11] Flask - a micro web framework written in Python. https://flask.palletsprojects.com/en/2.0.x/ (Accessed on December 25, 2021)

[12] Coral - build devices with local AI. https://coral.ai/ (Accessed on December 25, 2021)

[13] NVIDIA Jetson – Advanced AI Embedded Systems. https://www.nvidia.com/en-us/autonomous-machines/embedded-systems/ (Accessed on December 25, 2021)

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